M34 Racemic Epinephrine (Adrenaline)

Scope of Practice	EMT
Classification	• Adrenergic Agonist
Usual Presentation	• 0.5ml nebule containing 2.25% racemic epinephrine
Mechanism of Action	• Adrenaline acts on alpha-1 adrenergic receptors in the mucosal blood vessels of the larynx and trachea. This causes vasoconstriction, reducing blood flow to the inflamed tissues and decreasing edema (swelling) in the airway.
Indications	• Croup
Contraindications	• Known allergy to racemic epinephrine preparation.
Precautions	• Heart disease or structural abnormalities • Hypertension • Thyroid disease
Side Effects	• Tachycardia • Arrhythmias • Tremors • Arrhythmias (including VF) • Pulmonary edema • Headache
Adult Dose & Administration	• 0.5ml Nebulized (diluted in 3ml normal saline or sterile water)
Pediatric Dose & Administration	• **>4 years**: 0.5ml Nebulized (diluted in 3ml normal saline or sterile water) • **<4 years**: 0.05ml/kg (max 0.5ml, diluted in 3ml normal saline or sterile water)
Use in Pregnancy and Lactation	• Pregnancy: Safety has not been established. However racemic epinephrine should be administered when clinically indicated. • Lactation: Safety has not been established. However racemic epinephrine should be administered when clinically indicated.
Additional Information	• Nil.